

Week	Dates	Description	# Estimated Hours
Week 1	June 8 - June 14	Download TQC database (38,184 materials), JARVIS-DFT (1,868 topological materials), Materials Project via API. Download pre-trained model weights: DiffCSP, ALIGNN, M3GNet. Configure access to Yan lab HPC cluster. Perform EDA on data, Filter databases for light elements (Li-Si), and test that pretrained models perform as expected.	30
Week 2	June 15 - June 21	Generate baseline structures from unconditional DiffCSP. Evaluate baseline with topogivity, ALIGNN, M3GNet. Document baseline stability rates and topological hit rates. Checkpoint: Weekly meeting with Dr. Yan to review baseline performance.	30
Week 3	June 22 - June 28	Implement MatInvent PPO framework. Code multi-objective reward function combining topological probability, DOS at Fermi energy, formation energy, composition penalty. Debug reward computation pipeline. Set initial reward weights. Test RL training on small batch.	10
Week 4	June 29 - July 5	Run RL training iterations generating candidate structures. Monitor training curves: reward progression, KL divergence, policy gradient norms. Analyze reward distribution across objectives. Identify top candidates for checkpoint validation. Checkpoint: Weekly meeting to assess training dynamics.	25
Week 5	July 6 - July 12	Complete RL training. DFT-validate top checkpoint structures with VASP: relaxation + electronic structure. Compare GNN predictions to DFT ground truth. Retrain predictors if errors exceed thresholds. Adjust reward weights based on Pareto front analysis. Checkpoint: Review RL results with Prof. Yan, finalize candidate ranking.	15
Week 6	July 13 - July 19	Submit VASP jobs for top candidates (parallel execution): structure relaxation, electronic structure with SOC, band structure calculation. Compute topological invariants via Wilson loop analysis. Analyze formation energies, band gaps, density of states. Select materials meeting stability + topology criteria for phonon calculations.	15
Week 7	July 20 - July 26	Complete remaining VASP validations. Analyze electronic structures, identify materials with high DOS at Fermi energy. Set up Quantum ESPRESSO phonon calculations for most promising candidates. Submit phonon jobs (long-running, will complete during subsequent weeks).	10
Week 8	July 27 - Aug 2	Analyze phonon dispersion curves for dynamic stability (check for imaginary modes). Compute electron-phonon coupling strength λ from EPW code output. Estimate superconducting T_c via McMillan formula. Assess synthesizability via formation energy and structural prototype similarity to known materials. Checkpoint: Review validated materials with Prof. Yan.	15
Week 9	Aug 3 - Aug 9	Extract Pareto fronts from validated dataset (stability $\times$ topology $\times$ T_c -proxy). Generate 3D visualizations showing multi-objective optimization tradeoffs. Render crystal structures of top materials. Plot electronic band structures with topological features highlighted. Run statistical comparisons: RL vs. unconditional generation vs. random sampling.	15
Week 10	Aug 10 - Aug 16	Draft RISE presentation slides: introduction, methodology, results, implications. Prepare visualizations for general audience. Clean code for GitHub release: documentation, README, trained model weights, candidate database. Write 3-page technical summary. Final meeting with Prof. Yan: discuss Material Research Society submission, experimental synthesis feasibility.	15